

other.

D: The air particles oscillate parallel to direction of propagation.

18.

Ans: C

$$x = \frac{\lambda D}{a} \Rightarrow \lambda = \frac{ax}{D} = \frac{(0.1 \times 10^{-3})(8.0 \times 10^{-3})}{2.0} = 4.0 \times 10^{-7} \text{ m}$$

At the second order dark fringe, the path difference will be 1.5 times of the wavelength, $6.0 \times 10^{-7} \text{ m}$

19.

Ans: D